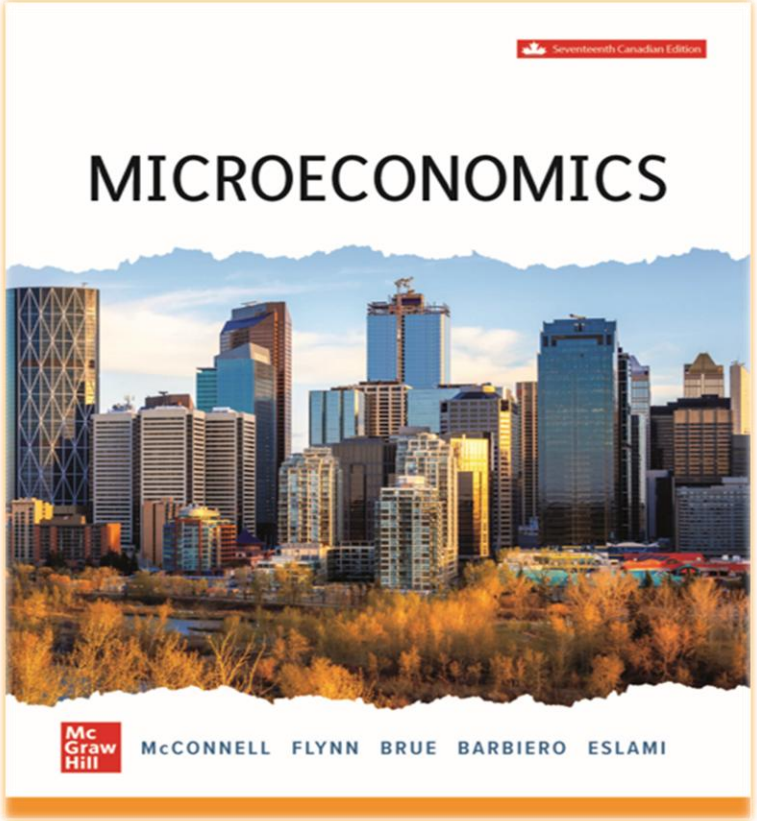
CHAPTER 5

Public Goods, Public Choice, and Government Failure



PowerPoint Presentation Prepared by:
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LEARNING OBJECTIVES

LO5.1 Describe free riding and public goods and illustrate why private firms cannot normally produce public goods.

LO5.2 Explain the difficulties of conveying economic preferences through majority voting.

LO5.3 Define government failure and explain its causes.

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INTRODUCTION

- Public goods can experience extreme market failure, leading to zero production, often requiring government intervention.
- The key economic issue is determining the allocatively efficient quantity of public goods the government should provide.
- Government failures, such as corruption, special interest lobbying, and voting paradoxes, can prevent optimal intervention.
- Both market failures and government failures require vigilance to ensure socially optimal outcomes.

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5.1 PUBLIC GOODS 1/9

Demand-side market failures

- When it is not possible to charge consumers for the product.
- Some can enjoy benefits without paying.
- Firms are not willing to produce since they cannot cover the costs.

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5.1 PUBLIC GOODS 2/9

Characteristics of Private Goods

- Majority of goods and services that are produced by private, for-profit companies.
- Characteristics
  - **Rivalry**
  - **Excludability**

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5.1 PUBLIC GOODS 3/9

Public Good Characteristics

- Provided by government
  - Offered for free
  - Characteristics
    - **Nonrivalry**
    - **Nonexcludability**
    - **Free-rider problem**

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Optimal Quantity of a Public Good

- Government estimates public good demand via surveys or votes since true demand isn't revealed in markets.
- Optimal provision is determined by comparing MB with MC.
- Following the MB = MC rule ensures efficient public good supply.

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5.1 PUBLIC GOODS 5/9

Demand for Public Goods

- Public good demand is unique, as it reflects the combined willingness to pay of all individuals in society.
- Economists estimate this demand through surveys assessing citizens' willingness to pay for different public goods.
- Table 5-1 shows the combined willingness to pay for national defence based on Garcia and Johnson’s marginal willingness to pay.

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TABLE 5-1 Demand for a Public Good, Two Individuals

(1) Quantity of Public Good	(2) Garcia's Willingness to Pay (Price)		(3) Johnson's Willingness to Pay (Price)		(4) Collective Willingness to Pay (Price)
1	$4	+	$5	=	$9
2	3	+	4	=	7
3	2	+	3	=	5
4	1	+	2	=	3
5	0	+	1	=	1

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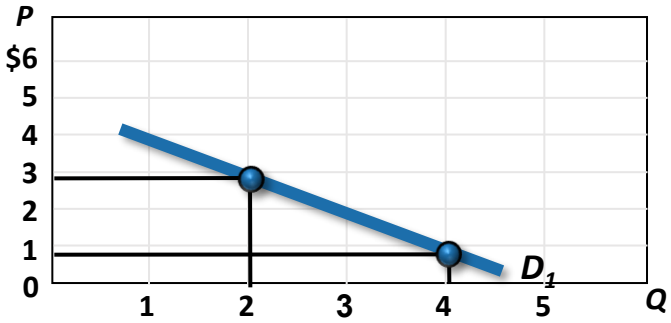
FIGURE 5-1 The Optimal Amount of a Public Good 1/3

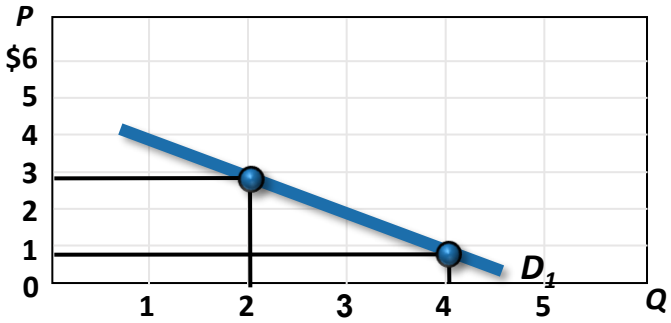
Garcia's Demand



(a) Garcia

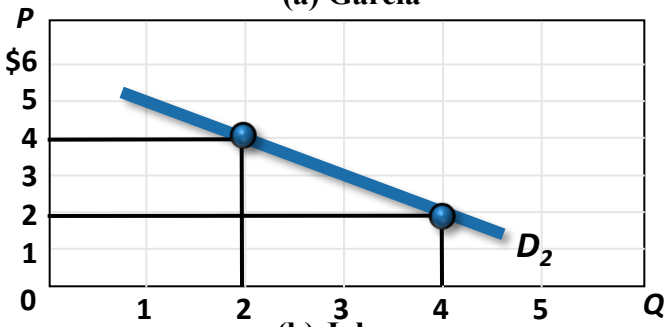
FIGURE 5-1 The Optimal Amount of a Public Good 2/3

Garcia's Demand



(a) Garcia

Johnson's Demand



(b) Johnson

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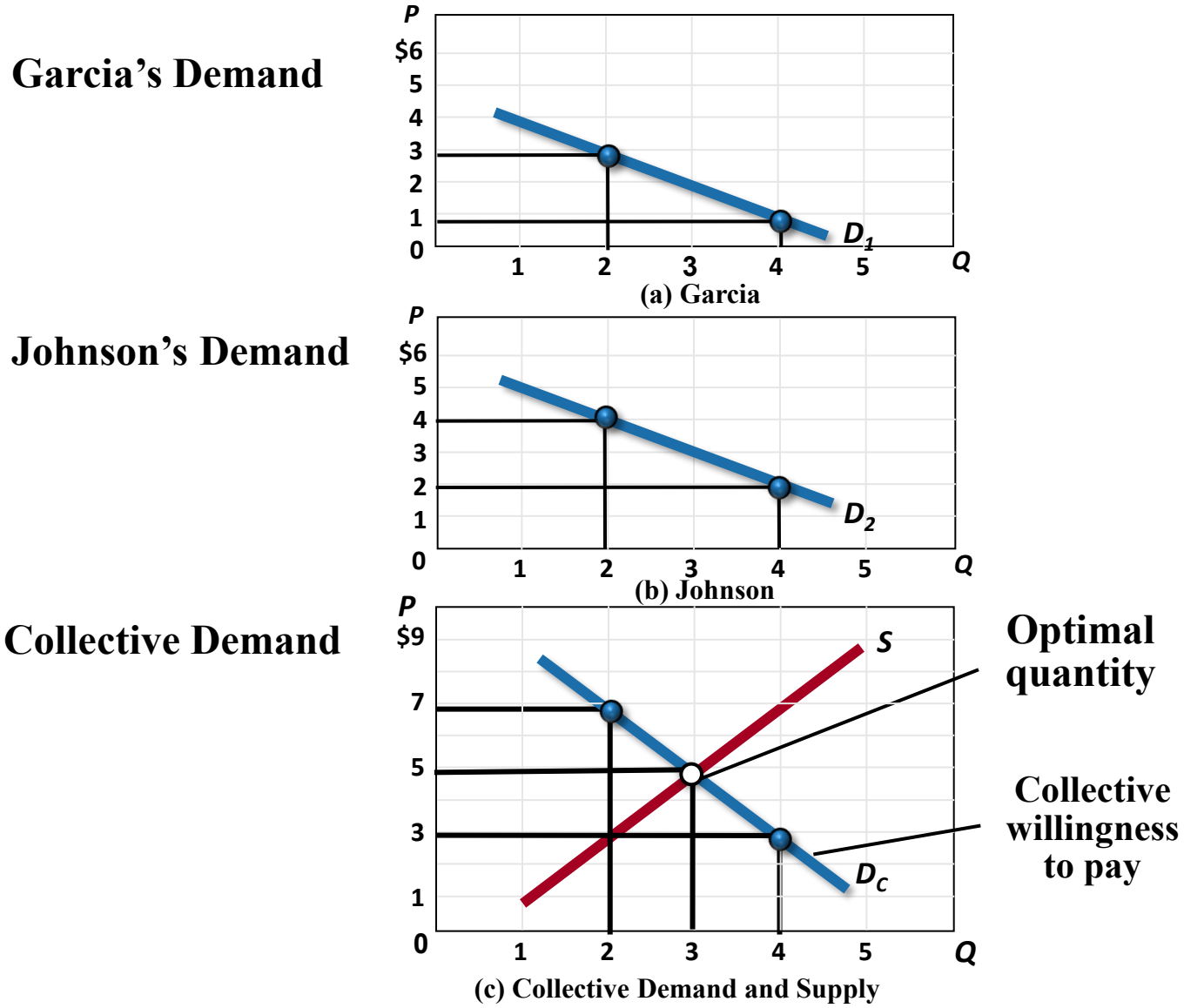
FIGURE 5-1 The Optimal Amount of a Public Good 3/3



5.1 PUBLIC GOODS 6/9

Comparing MB and MC

- The optimal quantity of a public good is where the collective demand curve (MB) intersects the supply curve (MC).
- In the example, the optimal quantity is three units, where the MB=MC.
- Equating MB and MC ensures efficient allocation of society's scarce resources.

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5.1 PUBLIC GOODS 7/9

Cost–Benefit Analysis

- Cost
  - Resources diverted from private goods production
  - Private goods that will not be produced
- Benefit
  - The extra satisfaction from the output of more public goods

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TABLE 5-2 Cost–Benefit Analysis for a National Highway Construction Project (billions of dollars) 1/6

(1) Plan	(2) Total Cost of Project	(3) Marginal Cost	(4) Total Benefit	(5) Marginal Benefit	(6) Net Benefit (4) – (2)
No new construction	$0	-	$0	-	$0
A: Widen existing highways	50	$50	200	$200	150
B: Two-lane highways	140	90	350	150	210
C: Four-lane highways	240	100	470	120	230
D: Six-lane highways	620	380	580	110	-40

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		MC		**MB**	

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TABLE 5-2 Cost–Benefit Analysis for a National Highway Construction Project (billions of dollars) 3/6

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MC < **MB**

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TABLE 5-2 Cost–Benefit Analysis for a National Highway Construction Project (billions of dollars) 4/6

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5.1 PUBLIC GOODS 8/9

Quasi-public Goods

- The government provides quasi-public goods like education and highways, which markets would underproduce.
- These goods offer broad societal benefits beyond individual buyers.
- Government steps in to prevent resource under allocation.

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5.1 PUBLIC GOODS 9/9

The Reallocation Process

- How are resources reallocated from the production of private goods to the production of public and quasi-public goods?
- Government
  - Taxes individuals and businesses
  - Takes the money and spends on the production of public goods

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5.2 PUBLIC CHOICE THEORY 1/15

Public Choice Theory

- Market failures like public goods and externalities justify government intervention, but government responses can also fail.
- Public choice theory explores the complexities of government decision-making and politics.
- Voting limitations can hinder government officials from determining public will and ensuring allocative efficiency.

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Revealing Preferences Through Majority Voting

- Determines the size and scope of government involvement in the economy
  - Reveals preferences
  - Results can be inconsistent and imperfect
    - **Presentation order of the choices**
    - **Distribution of the costs vs benefits**

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5.2 PUBLIC CHOICE THEORY 3/15

Inefficient Voting Outcomes

- Majority voting doesn’t always ensure the provision of such beneficial public goods.
- There can be a disconnect between the voting outcome and optimal resource allocation.
- This misalignment can result in suboptimal public good provision.

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Inefficient Voting Outcomes

INEFFICIENT "NO" VOTE

TB > TC

INEFFICIENT "YES" VOTE

TC > TB

- The government might fail to improve the economic inefficiency of overproduction or underproduction of specific public goods.

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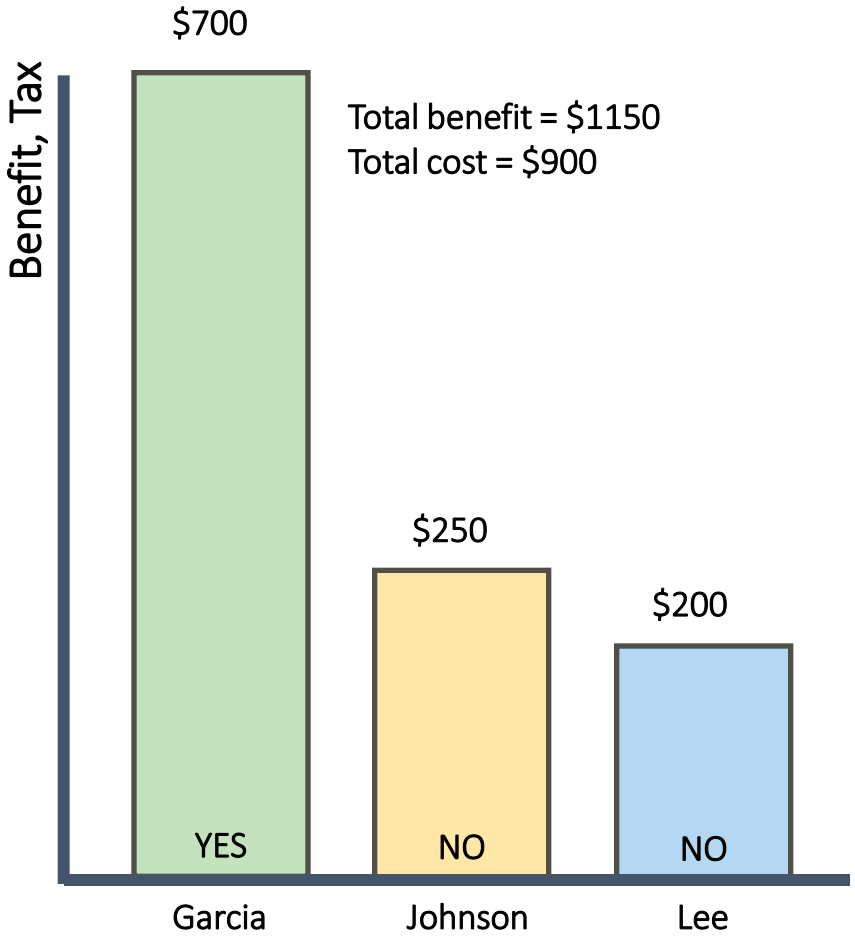
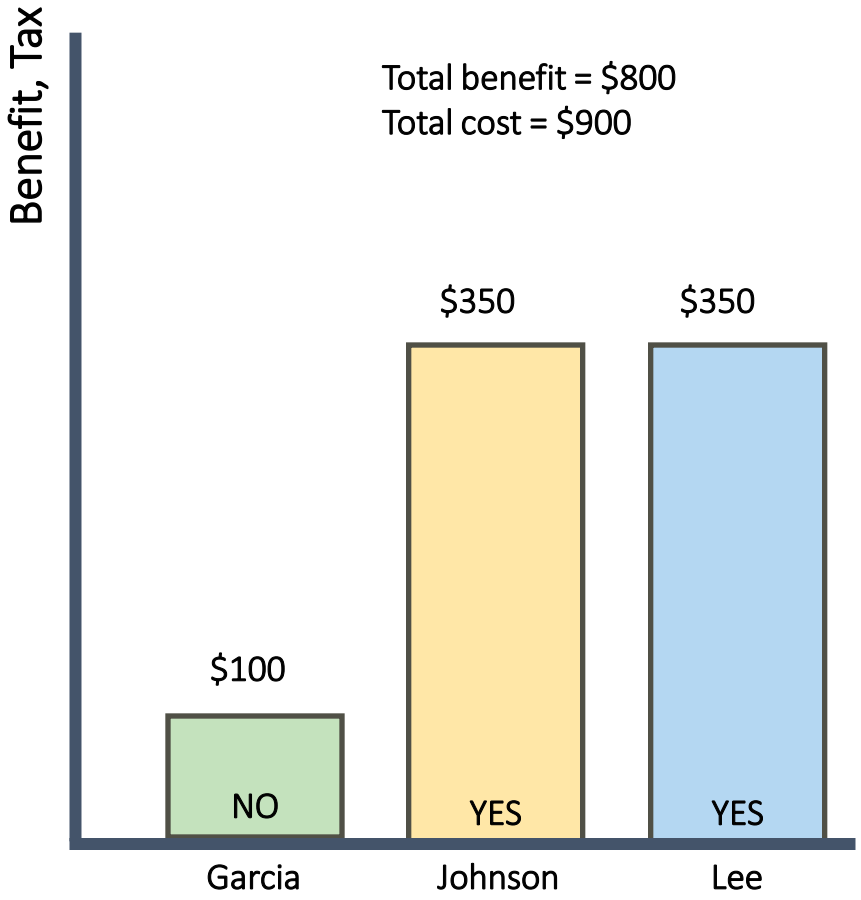
FIGURE 5-2 Inefficient Voting Outcomes 1/2



(a) Inefficient majority "no" vote



(b) Inefficient majority "yes" vote

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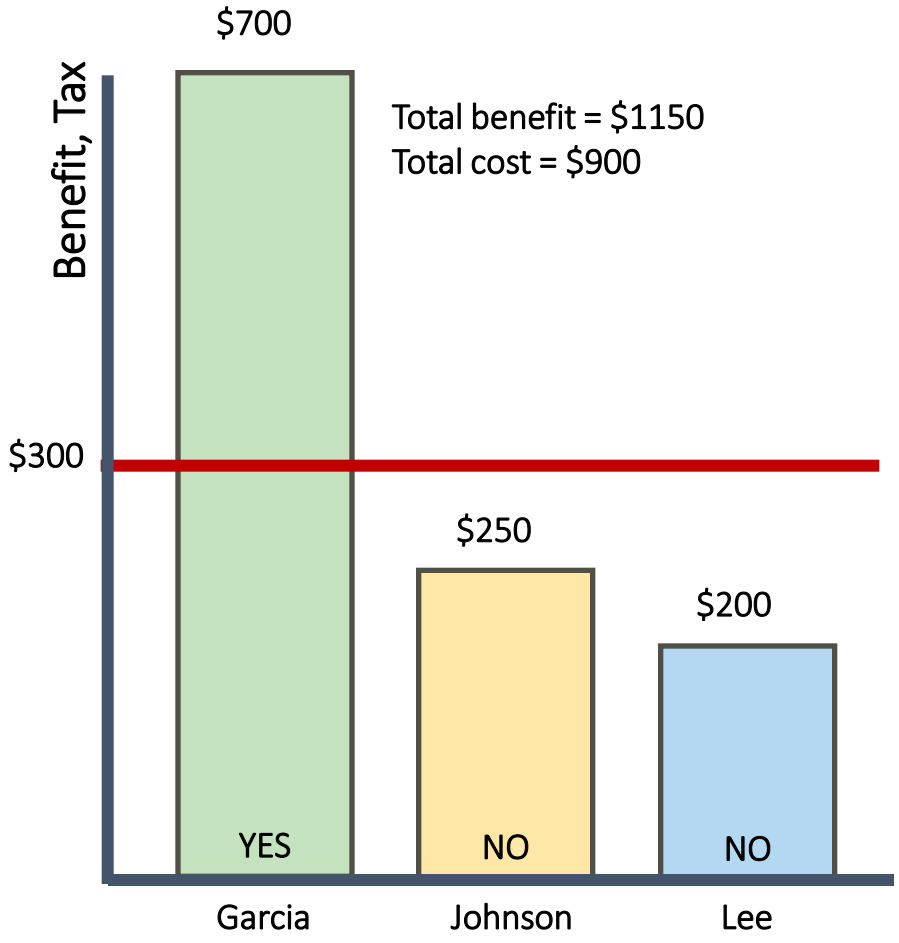
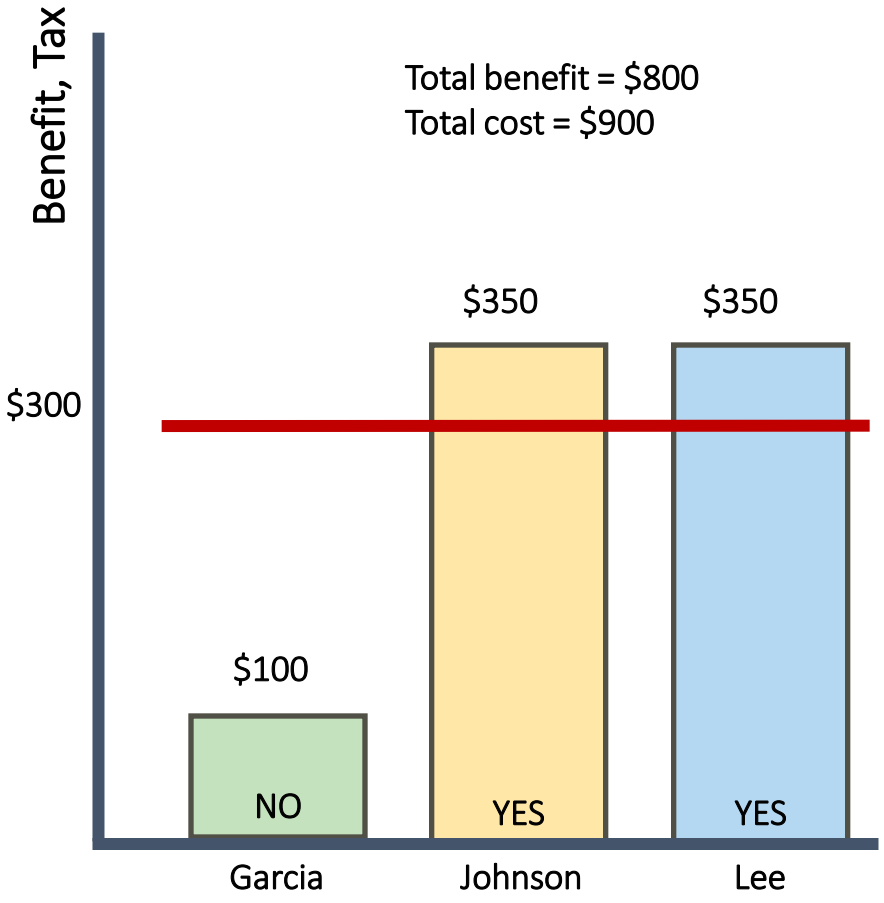
FIGURE 5-2 Inefficient Voting Outcomes 2/2



(a) Inefficient majority "no" vote



(b) Inefficient majority "yes" vote

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5.2 PUBLIC CHOICE THEORY 5/15

Implications

- Majority voting can lead to inefficient outcomes, causing either overproduction or underproduction of public goods.
- The government might fail to provide justifiable public goods while funding others not economically warranted.
- Majority voting doesn't account for the intensity of individual preferences, leading to potentially inefficient public good provision.

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Interest Groups And Logrolling

- **Interest groups**
  - Those who have a strong preference for a public good may band together.
- **Political logrolling**
  - The trading of votes to secure favourable outcomes.

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Paradox of Voting

- Majority voting can lead to the paradox of voting, where society struggles to rank preferences consistently.
- Paired-choice voting can produce inconsistent outcomes in ranking societal preferences.
- This inconsistency challenges the reliability of majority voting in decision-making.

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5.2 PUBLIC CHOICE THEORY 8/15

Preferences

- Consider Table 5-3. Three voters have different preferences for public goods: national defence, a road, and a weather warning system.
- Paired-choice majority voting can be used to determine the community's preferences by comparing two goods at a time.
- Individual preferences vary, with each voter ranking the three public goods in a distinct order.

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TABLE 5-3 Paradox of Voting

	Preferences		
Public Good	**Garcia**	**Johnson**	**Lee**
National defense	1st choice	3d choice	2d choice
Road	2d choice	1st choice	3d choice
Weather warning system	3d choice	2d choice	1st choice
Election	**Voting Outcomes: Winner**		
1. National defense vs. road	National defense (preferred by Garcia and Lee)		
2. Road vs. weather warning system	Road (preferred by Garcia and Johnson)		
3. National defense vs. weather waring system	Weather warning system (preferred by Johnson and Lee)		

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5.2 PUBLIC CHOICE THEORY 9/15

Voting Outcomes

- Paired-choice majority voting can produce inconsistent results, as shown by the community's conflicting preferences in Table 5-3.
- The outcome of majority votes can vary based on the voting sequence, leading to results that don't accurately reflect the electorate's true preferences.
- Although flawed, majority voting remains a better method for reflecting community preferences than alternatives like dictatorship or aristocracy.

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Median-Voter Model

- The median-voter model shows that in majority rule, the median voter decides election outcomes.
- The median voter holds a middle position, balancing stronger and weaker preferences on an issue.
- Extremes on both sides favor the median choice, making it the dominant outcome.

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Median-Voter Model: Real-World Applicability

- Public choices often align with the median view, as candidates adjust their positions to appeal to the median voter.
- Candidates first appeal to the median voter within their party for nomination, then shift towards the political center for the general election.
- Polls guide candidates in refining their positions to match the preferences of the broader median voter base.

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Median-Voter Model: Implications

1. The median-voter model implies that many people will be dissatisfied with the size of government, as it reflects only the median preference.
2. Individuals may relocate to areas where the median voter's preferences align more closely with their own, affecting their tax and service levels.

- Shifting preferences and imperfect information can cause politicians to misjudge the median position, impacting their electoral success.

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Alternative Voting Mechanisms

- These alternative systems allow voters to cast multiple votes, reflecting the intensity of their preferences.
- By considering preference strength, these methods aim to avoid inefficient voting outcomes.
- Alternative voting systems are more likely to approve public goods where the total benefit exceeds the total cost, leading to more efficient public good provision.

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Alternative Voting Mechanisms: Quadratic Voting

- Allows voters to purchase and cast multiple votes, reflecting the strength of their preferences.
- The cost of votes increases quadratically, encouraging voters to buy votes only up to where their marginal benefit equals the marginal cost.
- Aims to reduce inefficient outcomes by better aligning vote strength with the intensity of voter preferences.

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Alternative Voting Mechanisms: Mechanism Design

- Studies how to create rules or procedures, like quadratic voting, to optimize interactions and outcomes in various situations.
- Aim to develop voting systems that minimize the likelihood of inefficient votes.
- The effectiveness of a specific mechanism, such as quadratic voting, may vary depending on the context and scale, requiring careful consideration.

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5.3 GOVERNMENT FAILURE 1/10

- Government failure arises from inefficiencies in the public sector.
- Voting paradoxes, which obscure voter preferences, are one source of this failure.
- Government failure can also occur even when officials know exactly what citizens want.

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5.3 GOVERNMENT FAILURE 2/10

- **Representative Democracy and the Principal-Agent Problem**
- Arise when agents' interests conflict with those of the principals who delegate tasks to them.
- In business, this occurs when managers (agents) make decisions that benefit themselves but harm shareholders (principals).
- In politics, it leads to economic inefficiency, as politicians may prioritize re-election over constituents' best interests.

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5.3 GOVERNMENT FAILURE 3/10

Special-interest effect

- Small, organized groups can sway government policies in their favour against the majority's interests.
- They use campaign contributions and lobbyists to push their agenda.
- The group's benefits are concentrated, while costs are spread thinly across the majority, minimizing opposition.

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5.3 GOVERNMENT FAILURE 4/10

Special-interest effect: Pork Barrel Politics

- Politicians secure local projects to benefit their district.
- Local constituents gain, while costs are spread across a wider, less informed taxpayer base.
- Politicians pursue these projects to win favour with local voters.

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5.3 GOVERNMENT FAILURE 5/10

Rent-seeking behaviour

- Using government influence to secure payments or benefits beyond what is necessary to keep a resource employed.
- Influencing government policies for additional income.
  - Inefficient Examples: Tariffs, tax breaks, costly projects, excessive licensing, and subsidies reduce efficiency.

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5.3 GOVERNMENT FAILURE 6/10

Limited and Bundled Choice

- Voters and politicians face bundled choices, unlike selective purchasing in the marketplace.
- Voters must accept unwanted public goods to get desired ones, leading to inefficiency.
- Politicians must vote on entire packages, often including unrelated spending, reducing selectivity.

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5.3 GOVERNMENT FAILURE 7/10

Bureaucracy and Inefficiency

- Public agencies lack the profit-driven incentives for efficiency present in private businesses.
- Unlike private firms, public agencies have no clear measures like profit and loss to assess efficiency.
- Government employees and special-interest groups often block efforts to reduce or eliminate inefficient agencies.

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5.3 GOVERNMENT FAILURE 8/10

Inefficient Regulation and Intervention

- **Regulatory Capture**: Agencies may be influenced by the industries they regulate.
- **Deregulation**: Removing regulations can prevent capture but only works if competition thrives.
- Deregulation works in some industries, but regulation may still be needed in critical areas like safety and the environment.

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5.3 GOVERNMENT FAILURE 9/10

Corruption

- Political Corruption is the unlawful misdirection of governmental resources or actions for personal gain.
- Two basic forms of corruption:
  - Government official bribed to do part of his job.
  - Government official demands a bribe to do something illegal.

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5.3 GOVERNMENT FAILURE 10/10

Imperfect Institutions

- Criticisms of public-sector inefficiency challenge the idea of a perfectly benevolent government.
- Both markets and governments have imperfections, making it hard to decide which should handle certain activities.
- Goods like fire protection, housing, and education are often provided by both sectors, reflecting this complexity.

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THE LAST WORD 1/3

The COVID-19 Crisis in 2020 Showed Examples of Government Failure

- Canadian health authorities, confident from past SARS experience, underestimated COVID-19's severity.

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The COVID-19 Crisis in 2020 Showed Examples of Government Failure

- Canadian governments were less prepared for COVID-19 than anticipated, with significant shortcomings in long-term care, testing, PPE, and coordination.
- The federal government had to deploy the military to assist in overwhelmed care facilities, highlighting the lack of preparedness.

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The COVID-19 Crisis in 2020 Showed Examples of Government Failure

- Despite these issues, Canada's response was more cautious and effective than the U.S., leading to a lower death rate.
- By the end of 2020, Canada had 40 deaths per 100,000 people, compared to 101 in the U.S., a trend that continued throughout the pandemic.

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CHAPTER SUMMARY 1/3

- Free riding occurs when nonpayers benefit from public goods, which are nonrival and nonexcludable, making them unprofitable for private firms to produce.
- Governments can finance public goods through taxation, ensuring society's collective demand is met.
- Majority voting can lead to government failures, with potential underallocation or overallocation of resources and inconsistent voting outcomes.

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CHAPTER SUMMARY 2/3

- The median-voter model suggests that voters with the median preference on an issue determine election outcomes.
- Quadratic voting, where voters buy votes proportional to their preference strength, can lead to more efficient outcomes than traditional one-person-one-vote systems.
- Mechanism design in economics explores how to create rules that maximize the chances of achieving economically efficient outcomes.

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CHAPTER SUMMARY 3/3

- Government failure can result from principal-agent problems, the special-interest effect, and rent-seeking behaviour.
- Both government and markets have their imperfections, and it is crucial to understand where each is likely to succeed or fail in providing public goods.

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